

Week 4 — Persistence and Privilege Escalation

Overview

Week 4 focused on simulating post-exploitation persistence techniques. After gaining initial access in Week 3, an attacker's next goal is to ensure they can maintain access even after a system reboot, a password change, or an account lockout. We will use the local account of the victim considering that the attacker has already gotten control of it

Scenario	Tool	MITRE Technique
Scheduled Task Persistence	schtasks.exe	T1053
Registry Run Key Persistence	reg.exe	T1547
Backdoor Admin Account Creation	PowerShell	T1136, T1098

Pre-Week 4 Setup

Audit Policies Enabled

The following audit policies were enabled before running any attack scenarios to ensure Windows generates the required security log events.

```
auditpol /set /subcategory:"User Account Management" /success:enable  
/failure:enable  
auditpol /set /subcategory:"Process Creation" /success:enable  
auditpol /set /subcategory:"Other Object Access Events" /success:enable  
/failure:enable
```

Verified with:

```
auditpol /get /subcategory:"User Account Management"  
auditpol /get /subcategory:"Other Object Access Events"
```

Real World Note: Audit policies are disabled by default on Windows 10. This is a common misconfiguration in enterprise environments that leaves critical post-exploitation activity completely unlogged. Enabling and validating audit policy is a foundational step in any SOC deployment.

Scenario 1 — Scheduled Task Persistence

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Scheduled Task/Job	T1053

Objective

Simulate an attacker creating a scheduled task that executes a malicious payload on every user login, ensuring persistence after reboot or session termination.

Why Attackers Use This

Scheduled tasks are a native Windows feature that blend with legitimate system activity. They survive reboots, can run with elevated privileges, and are commonly overlooked during incident response if named convincingly.

Commands Executed on Windows

Create the scheduled task:

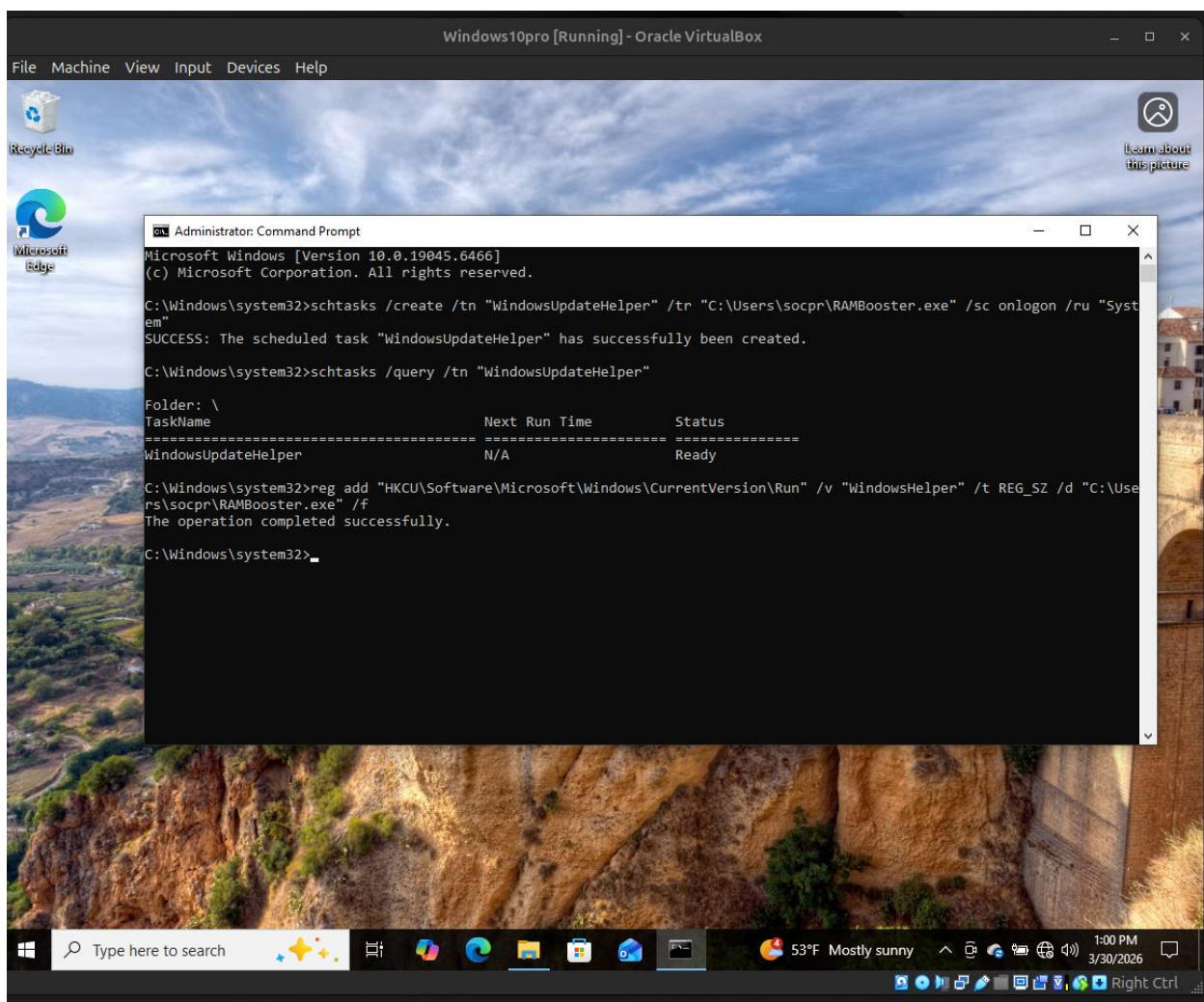


Fig 1: Creation of the Scheduled Task

Parameter breakdown:

Parameter	Value	Purpose
/tn	WindowsUpdateHelper	Task name disguised as legitimate
/tr	C:.exe	Payload from Week 3
/sc	onlogon	Triggers on every user login
/ru	System	Runs with SYSTEM privileges

Verify the task was created:

```
schtasks /query /tn "WindowsUpdateHelper"
```

Output:

TaskName	Next Run Time	Status
=====	=====	=====
WindowsUpdateHelper	N/A	Ready

Issue Encountered — EventCode 4698 Not Appearing

Problem: Initial Splunk search for EventCode 4698 returned no results.

Root Cause: The audit subcategory Other Object Access Events was not enabled, which is specifically required for scheduled task creation events to be written to the Security log.

Resolution:

```
auditpol /set /subcategory:"Other Object Access Events" /success:enable  
/failure:enable
```

The task was deleted and recreated after enabling the policy. EventCode 4698 then appeared in Splunk.

Fallback Detection Used: Sysmon EventID 1 was also used as a fallback to detect schtasks.exe process execution regardless of audit policy state. This demonstrates why layered telemetry across multiple sources is critical.

Logs Generated

Event ID	Source	Description
4698	Windows Security	Scheduled task created
4699	Windows Security	Scheduled task deleted
Sysmon 1	Sysmon	schtasks.exe process creation

Splunk Detection Queries

Detect scheduled task creation:



Fig 2: Detection of a Scheduled Task being created

Dashboard Results

The SOC_Week_4 dashboard confirmed:

- Scheduled Task Creation Over Time timechart showing a spike at 21:29 on 2026-03-30
- Scheduled Tasks Created panel showing the event on DESKTOP-6C9QB2R at 2026-03-30 21:29:11
- Full Persistence Activity Timeline showing Scheduled Task Created at 21:29:11

Scenario 2 — Registry Run Key Persistence

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Boot/Logon Autostart Execution: Registry Run Keys	T1547

Objective

Simulate an attacker modifying the Windows registry to execute a payload automatically on every user login via the Run key autostart mechanism.

Why Attackers Use This

Registry Run keys are one of the oldest and most common persistence mechanisms. They execute automatically on login, require no elevated privileges for the HKCU hive, and blend in alongside legitimate software autostart entries such as OneDrive. We will call it WindowsHelper to make it seem legitimate

Commands Executed on Windows

Add the registry Run key:

Logs Generated

Event ID	Source	Description
Sysmon 1	Sysmon	reg.exe process creation
Sysmon 13	Sysmon	Registry value set — WindowsHelper key written
4657	Windows Security	Registry value modified (requires object access auditing)

Splunk Detection Queries

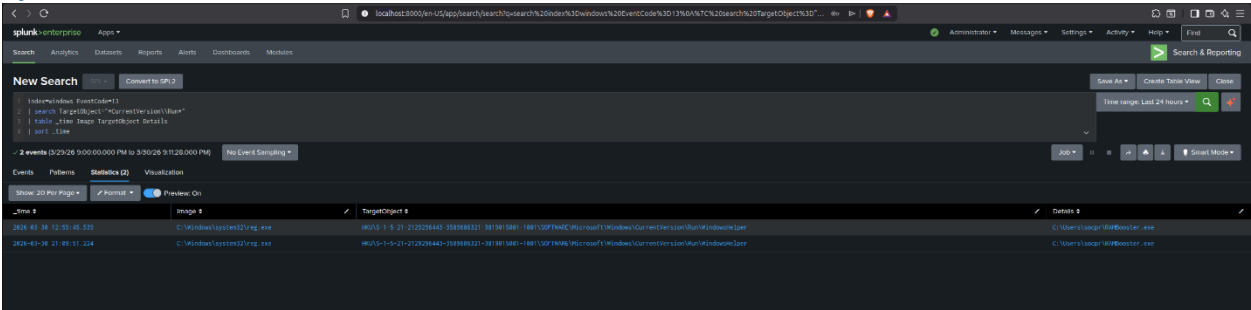


Fig 5: Clean Splunk Query

Scenario 3 — Backdoor Admin Account Creation

MITRE ATT&CK Mapping

Tactic	Technique	ID
Persistence	Create Account: Local Account	T1136
Privilege Escalation	Account Manipulation	T1098

Objective

Simulate an attacker creating a backdoor administrator account to maintain persistent access even if the original compromised credentials are changed or the account is disabled.

Why Attackers Use This

A secondary admin account named to resemble a legitimate service account provides a reliable fallback access method. It survives password resets on the primary account and is easy to overlook during a casual user account review.

Commands Executed on Windows

Create the backdoor account:

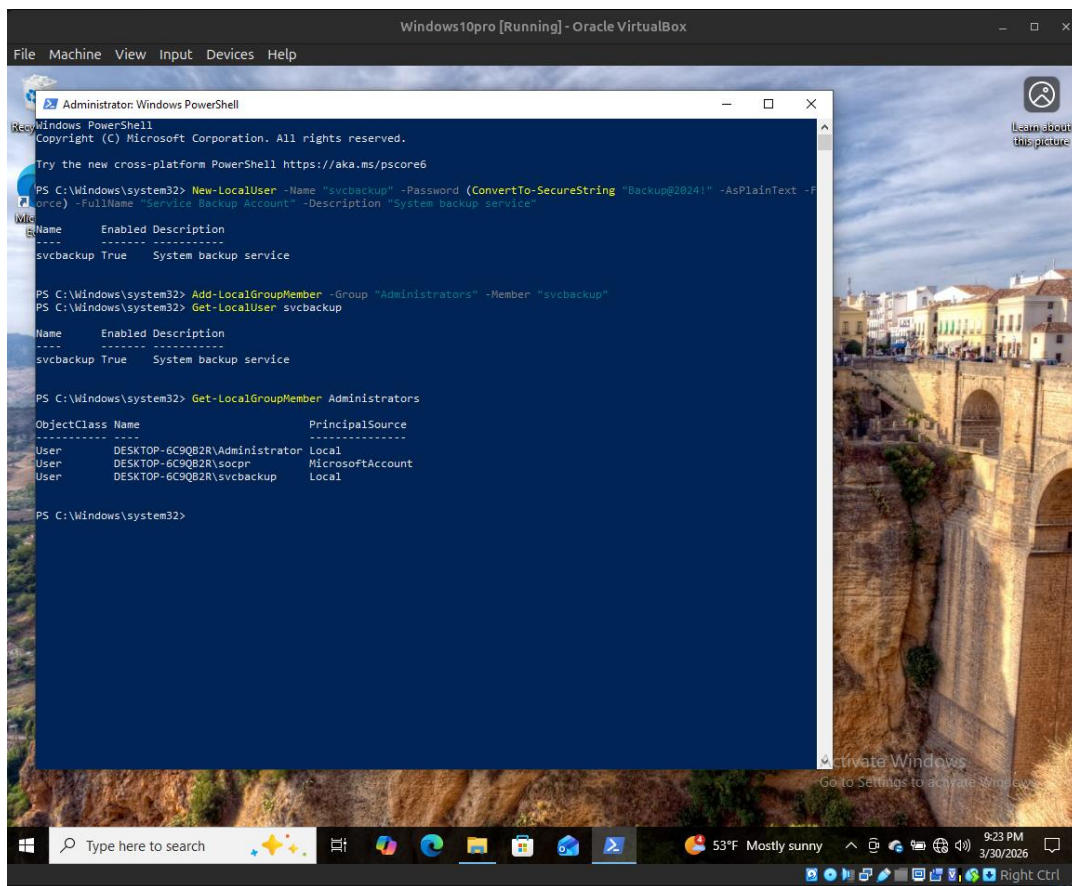


Fig 6: Creation of an Administrator Account

Account Naming Strategy

The name `svcbackup` was chosen deliberately to mimic a legitimate backup service account. The description “System backup service” adds further legitimacy. This demonstrates how attackers use naming conventions to avoid detection during casual account reviews.

Logs Generated

Event ID	Source	Description
4720	Windows Security	User account <code>svcbackup</code> created
4722	Windows Security	User account <code>svcbackup</code> enabled
4732	Windows Security	<code>svcbackup</code> added to local Administrators group

Splunk Detection Queries

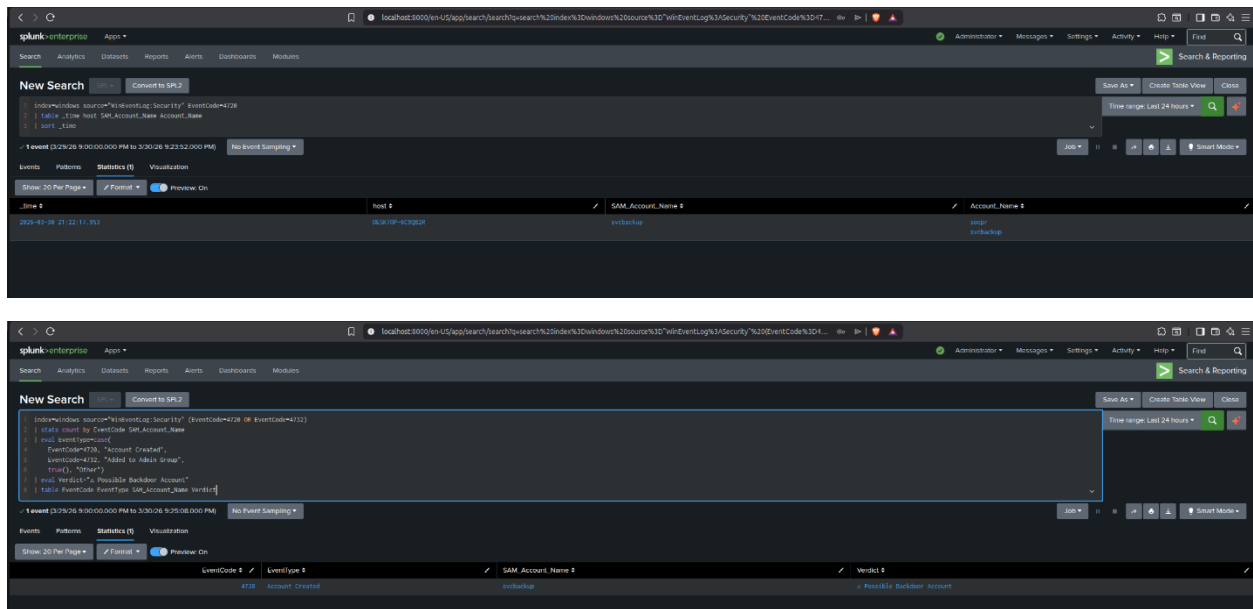


Fig 7: Detection of a Scheduled Task being created

Dashboard Results

The SOC_Week_4 dashboard confirmed:

- New User Accounts Created panel: svcbackup on DESKTOP-6C9QB2R at 2026-03-26 21:22:17
- Full Persistence Activity Timeline showing Account Created at 21:22:17 and Scheduled Task Created at 21:29:11 in the same session

Cumulative Dashboard for Week 4

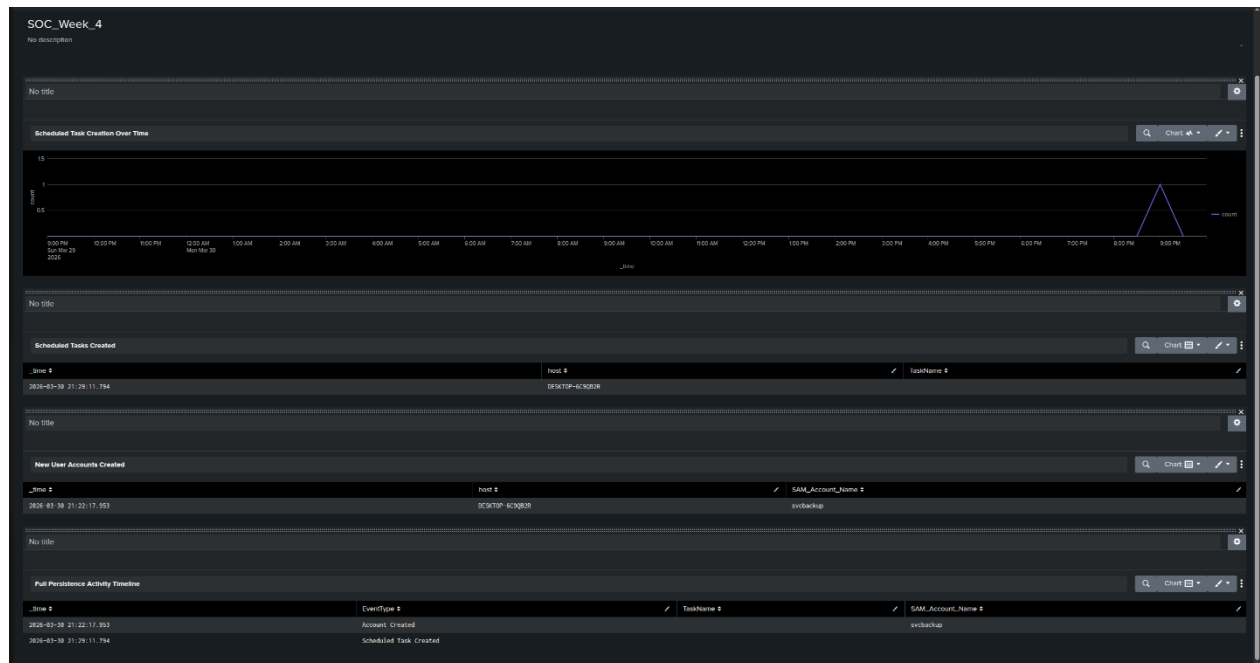


Fig 8: Week 4 Dashboard

Week 4 Summary — Issues and Resolutions

Issue	Root Cause	Resolution
EventCode 4698 not appearing	Other Object Access Events audit policy disabled	Enabled via auditpol command
Registry search returning 75+ noise events	Normal Windows background process activity	Filtered by TargetObject and excluded svchost.exe
Registry detection rule too broad	Run and RunOnce search matching system processes	Removed broad rule, replaced with targeted queries

MITRE ATT&CK Coverage — Week 4

Attack	Tactic	Technique	ID
Scheduled task	Persistence	Scheduled Task/Job	T1053
Registry run key	Persistence	Boot/Logon Autostart Execution	T1547
Backdoor account creation	Persistence	Create Account: Local Account	T1136
Admin group addition	Privilege Escalation	Account Manipulation	T1098

Key Lessons Learned

1. Audit policy gaps create detection blind spots. EventCode 4698 for scheduled task creation requires a specific audit subcategory that is not enabled by default. Without it, scheduled task persistence is completely invisible in the Security log. This misconfiguration is common in real enterprise environments.

2. Layered telemetry provides resilience. When EventCode 4698 was initially missing, Sysmon EventID 1 still detected schtasks.exe execution. Multiple telemetry sources ensure that no single policy misconfiguration creates a complete blind spot.

3. Registry telemetry is extremely noisy. Windows generates over 400 registry modification events per minute during normal operation. Broad detection rules produce too many false positives to be actionable. Precision filtering by TargetObject and process exclusion is essential.

4. Attacker naming conventions are deliberate. Both WindowsUpdateHelper and svcbackup were named specifically to blend with legitimate system activity. SOC analysts must baseline normal scheduled task names and local account names to identify anomalies effectively.

5. The full lifecycle of an attack matters. Capturing both task creation (4698) and task deletion (4699) demonstrates the complete attack and cleanup cycle. Real attackers often attempt to clean up evidence after establishing persistence.